Cloud Architecture

Types of cloud computing

Summary Notes

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Lesson objectives

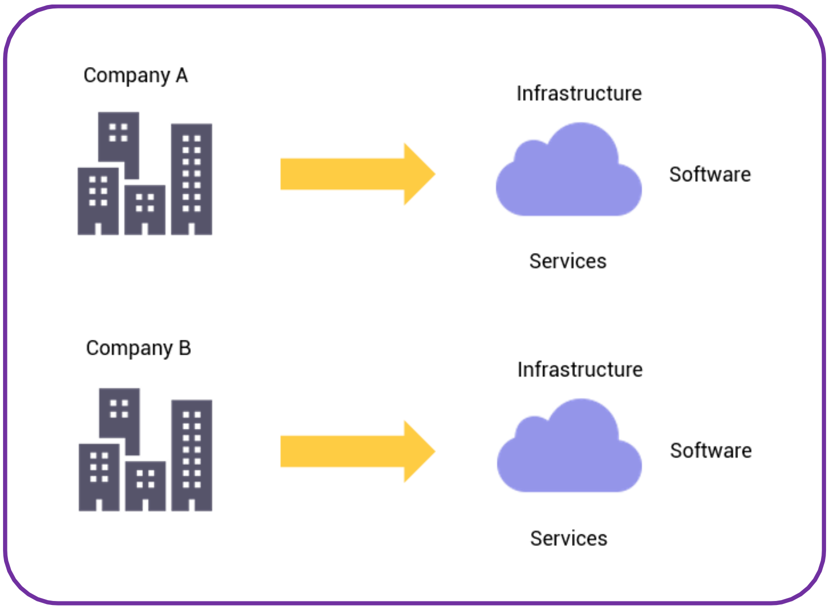
By the end of this lesson, you should be able to:

- Distinguish between the types of cloud computing
- Explore the architectures of cloud computing types
- Identify security considerations

Types of cloud computing

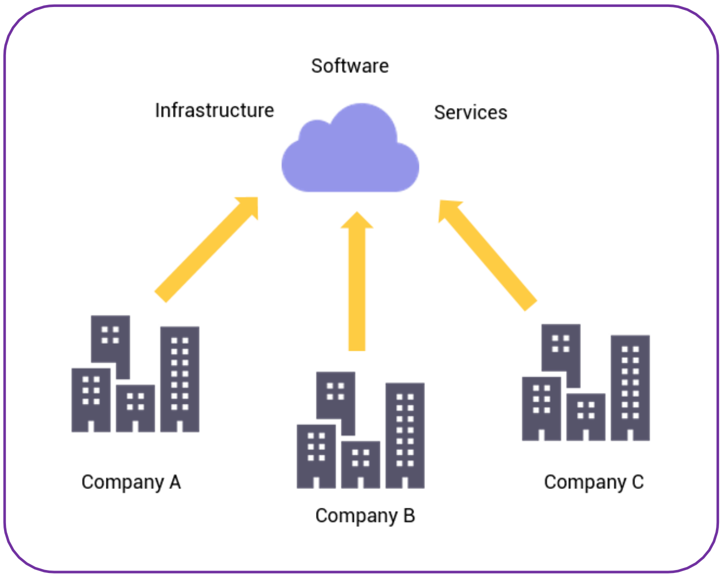
Private clouds

Private clouds are owned and managed by single private entities such as organisations. The cloud infrastructure is usually hosted in a central physical location within the organisation – within a data centre. However, there are exceptions where an organisation might deploy and manage the cloud while offloading a fraction of administrative services to a third-party company.



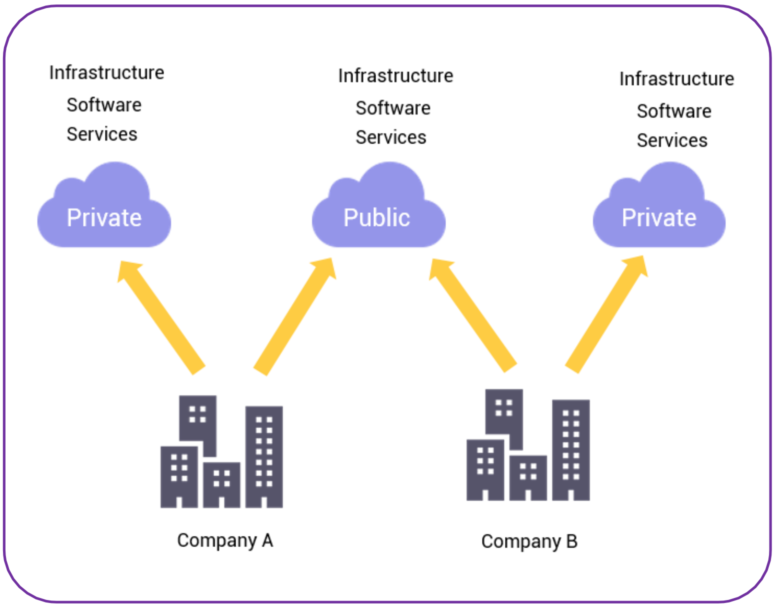
Public clouds

The public cloud refers to cloud computing services that are made available to the general public over the internet. These services are owned and operated by third-party companies, such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP), and are available on a pay-as-you-go basis. Public cloud services can be used for a wide range of purposes, including storing and processing data, hosting websites and web applications, and running software and applications.



Hybrid clouds

A hybrid cloud is a type of cloud computing that combines a public cloud and a private cloud. A hybrid cloud allows an organization to use the public cloud for certain workloads and the private cloud for others, depending on the specific requirements and constraints of each workload. For example, an organisation may choose to use the public cloud for running web applications and storing data, while keeping sensitive data and applications on the private cloud.



Cloud migration considerations

- **Security**: One of the most important considerations is ensuring that data is secure while it is stored and transmitted in the cloud. This includes protecting against data breaches, unauthorised access, and data loss.
- **Compliance**: Organisations need to ensure that they are in compliance with any industry-specific regulations or laws that apply to their data and applications.
- **Latency and performance**: Cloud providers may have data centres located in different regions, so it's important to consider the performance impact of hosting data and applications in different geographical locations.
- **Data sovereignty**: If an organization is subject to laws and regulations that restrict where data can be stored and processed, this may need to be considered when choosing a cloud provider.